



**Verified Carbon
Standard**

HYDROELECTRIC PROJECT EL EDÉN

Project proponent

Central Hidroeléctrica El Edén S.A.S. E.S.P.

Document Prepared by ALLCOT AG

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1 PROJECT DETAILS

1.1 Summary Description of the Implementation Status of the Project

The Hydroelectric Project El Edén (hereafter, the project activity or project) is implemented by Central Hidroeléctrica El Edén S.A.S E.S.P (hereafter, the project proponent) and has been verting electricity to the National Interconnected System (SIN, for its initials in Spanish) since the 16-February-2017¹.

The hydroelectric power plant utilizes water from the river “La Miel”, located in the river basin “La Miel” in the east of the Department of Caldas, Colombia (hereafter, the Host Country), about 125 kilometres from the City of Cali and close to the municipalities of Pensilvania, Marquetalia and Manzanares.

The project activity is a run-of-river hydroelectric plant that includes the construction and installation of a concrete weir and intake, a desander, channel, forebay/settling tank, high-pressure penstock, powerhouse and discharge channel. Two Pelton turbines with 10.319 MW and two synchronous generators with 11,154 kVA each. Each generator will be connected to a transformer to increase the voltage from 13.2 kV to 33 kV at the project site. The installed capacity of the plant at the interconnection is 19.9 MW, as per the document “Certificado de Cumplimiento de Código de Conexión”.

The most relevant implementation dates of the project are shown below:

- 28-June-2013: Signing of the *Contract for the supply, assembly, and commissioning of electromechanical equipment*.
- 4-November-2016: Request for registration in the Public Service Providers Registry (RUPS – for its translation in Spanish) to the Superintendence of Public Domiciliary Services.
- 22-November-2016: Approval of the request for registration in the RUPS.
- 28-February-2017: End of Commissioning Certificate.
- 28-February-2017: Certificate of compliance with the Connection Code (Resolution CREG 025 of 1995).

The net electricity generated by the project during the monitoring period 16-February-2017 to 31-December-2020 was 237,996 MWh which was delivered to the National Interconnected System (SIN), thus increasing the share of renewable energy and reducing a total of 73,110 tCO₂ during the same period.

¹ According to the SIN administrator, XM. Data available at:
<http://portalbisrs.xm.com.co/oferta/Paginas/Despacho/genrealporage.aspx>

1.2 Sectoral Scope and Project Type

Scope 1 - Energy (renewable/non-renewable)

The project consists of only one hydroelectric power plant and is not a grouped project.

1.3 Project Proponent

The project proponent is Central Hidroeléctrica El Edén S.A.S E.S.P, which will be responsible for the construction, operation and maintenance of the project. Contact information is shown below.

Organization name	<i>Central Hidroeléctrica El Edén S.A.S. E.S.P.</i>
Contact person	<i>William Andres Bravo Horta</i>
Title	<i>Director of Generation</i>
Address	<i>Finca el Edén Km 2. vía Manzanares-Pensilvania.</i>
Telephone	<i>+(57) 3206909362</i>
Email	<i>-</i>

Organization name	<i>Central Hidroeléctrica El Edén S.A.S. E.S.P.</i>
Contact person	<i>Paula Andrea Villegas Londoño</i>
Title	<i>Legal Representative</i>
Address	<i>Calle 100 No. 13-21 Piso 15, Edif. Mega Tower</i>
Telephone	<i>(+571) 7455060 - Ext. 570</i>
Email	<i>-</i>

1.4 Other Entities Involved in the Project

Organization name	<i>ALLCOT AG</i>
Role in the Project	<i>Consultant</i>
Contact person	<i>Alexis Leroy</i>
Title	<i>CEO</i>
Address	<i>Steinhauserstrasse, CH-6300, Zug, Switzerland</i>

Telephone	(57) 322 209 8737
Email	ceo@allcot.com

1.5 Project Start Date

The project started operation on the 16-February-2017², which is the project start date. This date has been adjusted since the initial starting date was expected to be on 01-January-2015³.

1.6 Project Crediting Period

The first crediting period is 10 years, from 16-February-2017, date in which the project began to inject electricity to the SIN and therefore generating GHG emission reductions, to 15/February/2027. This date has been adjusted since the initial crediting period start date was expected to be on 1/September/2015 according to the registered VCS PD.

1.7 Project Location

The project is located in the Department of Caldas, Colombia. The project area includes the municipalities of Pensilvania, Marquetalia and Manzanares.

The coordinates of the intake are E 885038, N 1079832 (datum Bogotá). The following figures show the location of the project.

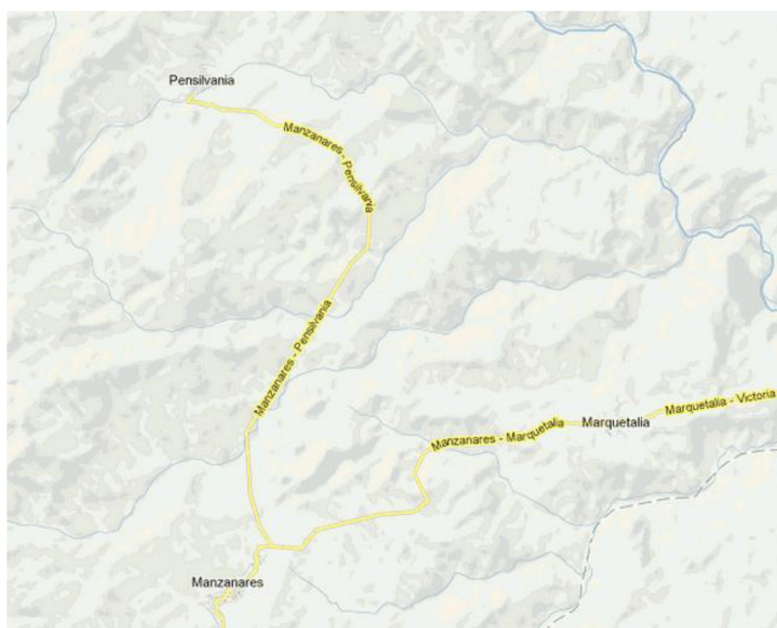


Figure 1. Project location in Colombia

² According to the SIN administrator, XM. Data available at: <http://portalbissrs.xm.com.co/oferta/Paginas/Despacho/genrealporage.aspx>

³ For more detail on this adjustment, please refer to section 3.2.2 Project Description Deviations of this Monitoring Report.

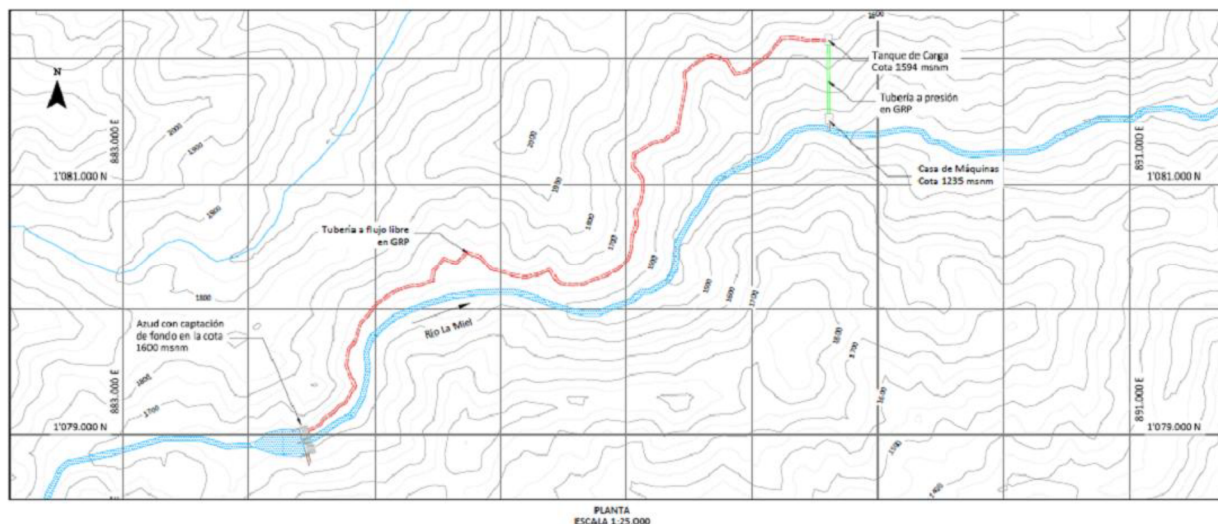


Figure 2. Project location

1.8 Title and Reference of Methodology

The project activity is developed in accordance with the approved consolidated CDM baseline and monitoring methodology ACM0002 “Consolidated baseline methodology for grid-connected electricity generation from renewable sources” (version 13.0.0).

(available at: <https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG>)

The identification of the baseline scenario and the demonstration of additionality are assessed by applying the latest versions of the CDM TOOL01 “Tool for the demonstration and assessment of additionality” (version 7.0.0).

(available at: https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-01-v7.0.0.pdf/history_view)

The emission factor of the relevant power-grid is determined based on the procedures of the CDM TOOL07 “Tool to calculate the emission factor for an electricity system” (version 3.0.0).

(available at: https://cdm.unfccc.int/methodologies/PAMethodologies/tools/am-tool-07-v7.0.pdf/history_view)

Moreover, the following guidelines are applied:

“Guidelines on the assessment of investment analysis” (version 5) as per the Annex 5 of CDM EB 62 report.

(available at: https://cdm.unfccc.int/Reference/Guidclarif/reg/reg_guid03.pdf)

“Guidelines on common practice” (version 02.0) as per the Annex 8 of CDM EB 69 report.

(available at: https://cdm.unfccc.int/Reference/Guidclarif/meth/meth_guid44.pdf)

1.9 Participation under other GHG Programs

The project has not been registered and is not seeking registration under any other GHG programs.

1.10 Other Forms of Credit

The project has no intends to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program.

1.11 Sustainable Development

The project activity facilitates the achievement of electricity demand required for the social and economic development of the country based on a renewable energy source. The project contributes to the region's sustainable development in the following ways:

- The project decreases emissions of sulphur oxides (SO_x), nitrogen oxides (NO_x), carbon monoxide, particulate matter, other pollutants, as well as carbon dioxide associated with the combustion of fossil fuels.
- The project reduces the dependence on fossil fuels, which are non-renewable and limited in supply.
- During construction, the project provided employment opportunities. Besides, employment has also been generated for operation and maintenance activities.

2 SAFEGUARDS

2.1 No Net Harm

The environmental impacts anticipated in the Environmental Impact Assessment (EIA) during the project construction and operation phase, have been managed through the implementation of the measures established in the Environmental Management Plan (EMP), as well as the Operation and Maintenance Plan (O&MP).

As noted in the VCS PD, the conclusions of the EIA were positive and allowed the issuance of the Environmental License on May 4,. A small modification of the project design was made subsequently, upon which the modified EIA was presented, and a second environmental license was released on August 5, 2013.

During the past few years (2014, 2018, 2020), other modifications have been requested and approved by the environmental authority. The most recent was issued on September 16, 2020.

In orther to demonstrate compliance of the Environmental License and its' environmental, operation and maintenance plans, Compliance Reports have been presented to Corpocaldas, the local environmental authority. Each report includes description of the project, maintenance activities, personnel responsible for the environmental activities, updated schedules and stages of the environmental and social plans and programs, as well as environmental compliances forms requested by Corpocaldas.

2.2 Local Stakeholder Consultation

The local stake holder consultation is part of the Environmental Impact Assessment (EIA) and its Environmental Management Plan (EMP), and Operation and Maintenance Plan (O&MP), according to the Colombian regulation.

Therefore, the Compliance Reports mentioned in the previous section also include the follow up of the six social programs stated in the EMP and O&MP, whose objectives are described below:

PMA-MS-01. Community and institutional information and participation: Evaluate the understanding of the information provided about the hydroelectric project and its characteristics during the different stages of the project.

PMA-MS-02. Information and training for personnel linked to the project: Track the tasks related to environmental responsibility, assigned to all personnel.

PMA-MS-03. Support to the institutional management capacity: Monitor the development of activities that aim to strengthen the capacity for action and management of public entities and organized groups.

PMA-MS-04. Information and sensibilization of the project to the local community: Inform the population that lives within the area of influence, about the activities of the project, during the studies, construction, and operation phases.

PMA-MS-05. Local labor hiring program: Evaluate the delivery of information on employment to the communities and authorities of the area of influence and look for employment opportunities to be directed mostly to personnel in the area.

PMA-MS-06. Impact on third parties: Follow up on the requirements made by the populations within area of influence, about the Hydropower.

Each one of these programs, have their own procedures / methodologies, indicators, goals and results which can be seen in detail in the Compliance Reports.

2.3 AFOLU-Specific Safeguards

This is non AFOLU project, therefore AFOLU-Specific Safeguards does not apply.

3 IMPLEMENTATION STATUS

3.1 Implementation Status of the Project Activity

The Hydroelectric Project El Edén has been operating since 16-February-2017, and no mayor breakdowns have happened. The project continues to operate under normal conditions.

However, some observations of the daily reports show turbidity and /or contamination on the incoming water flow. Sometimes, even obstructing the pipe for measuring the flow.

There have been some short stops during the operation mainly due to electrical storms, low level of the river or mechanical issues. Most of these stops have been quickly solved and operation restored within a few minutes.

Other partial but longer stops of the equipment have taken place for routine maintenance and have usually lasted for a few hours.

Finally, for stops related to the electrical equipment, the equipment provider has been contacted in order to solve the issue and operation has been restored.

3.2 Deviations

3.2.1 Methodology Deviations

No methodology deviations have been necessary for this Monitoring Report; therefore, this section does not apply.

3.2.2 Project Description Deviations

The main project description deviation for Hydroelectric El Edén is on the starting operation date which according to the VCS PD was supposed to be on the 01-January-2015 and ended up being on the 16-February-2017.

The expected construction date according to the VCS PD was June 2013 and the actual signing of the Contract for the supply, assembly, and commissioning of electromechanical equipment happened on the 28/June/2013, therefore, the construction started just as planned in the VCS PD. However, there were some difficulties during the construction phase, mainly on the excavation and underground activities, causing a delay on the starting operation date. The End of Commissioning Certificate was signed on the 28/February/2017. However, according to the SIN administrator, XM, the project Hydroelectric El Edén has been verting electricity to the SIN, on the 16-February-2017.

Thus, there is also a deviation on the project crediting period which was meant to start on the 01/09/2015. However, since the project only started generated GHG emissions reductions on the 16-February-2017, the project crediting period is 10 years, from 16-February-2017, date in which the project began to inject electricity to the SIN and therefore generating GHG emission reductions, to 15/February/2027.

There were also, some minor changes in the main equipment capacity (turbines, generators and transformers). However, it was stated in the VCS PD that the values proposed then, were based on the designs, and could vary slightly at real conditions. The capacity of the actual installed equipment is shown below:

- Turbines: 10,319 kW
- Generators: 11,154 kVA
- Transformers: 12,000 kVA

It was also stated in the VCS PD that the installed capacity of the plant at the interconnection point will be around 19.5 MW. According to the Certificate of compliance with the Connection Code (Resolution CREG 025 of 1995) issued by the grid operator, the installed capacity is 19.9 MW⁴.

3.3 Grouped Projects

This is non-Grouped project; therefore, this section does not apply.

4 DATA AND PARAMETERS

4.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using version 3.0.0 of the “Tool to calculate the emission factor for an electricity system”
Source of data	Calculated based on information provided by the grid operator XM.
Value applied	0.3072 tCO ₂ /MWh.
Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” (version 3.0.0).
Purpose of Data	Calculation of baseline emissions
Comments	--

4.2 Data and Parameters Monitored

Data / Parameter	EG _{facility,y}
Data unit	MWh/yr
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Electricity meter(s)

⁴ as per the document “Certificado de Cumplimiento de Código de Conexión”.

Description of measurement methods and procedures to be applied	The following parameters are measured: The meters installed are bi-directional, taking into account both the quantity of electricity supplied by the project plant to the grid and the quantity of electricity delivered to the project plant from the grid. In this case the individual values are not required.																				
Frequency of monitoring/recording	Continuous measurement and at least monthly recording. Typically, the measured data is read once every 24 hours using tele-metetering technology (remotely) and reported to XM (grid operator and administrator).																				
Value monitored	<table><tr><th>Year</th><th>E_g_{facility,y} (MWh)</th></tr><tr><td>2017</td><td>66,783.62</td></tr><tr><td>2018</td><td>78,259.22</td></tr><tr><td>2019</td><td>48,055.58</td></tr><tr><td>2020</td><td>44,898.41</td></tr><tr><td>Total</td><td>237,996.83</td></tr></table>	Year	E _g _{facility,y} (MWh)	2017	66,783.62	2018	78,259.22	2019	48,055.58	2020	44,898.41	Total	237,996.83								
Year	E _g _{facility,y} (MWh)																				
2017	66,783.62																				
2018	78,259.22																				
2019	48,055.58																				
2020	44,898.41																				
Total	237,996.83																				
Monitoring equipment	A main and a backup meter were installed, both bi-directional and with the following characteristics for both meters: Brand: Schneider Electric Model: Power Logic ION8650 socket meter (2011) Accuracy class: 0.2S Digital fault recording: Simultaneous capture of voltage and current channels for sub-cycle disturbance transients. Main meter serial number: MW-1509A351-02 Backup meter serial number: MW-1509A352-02																				
QA/QC procedures to be applied	Calibration was done according to the internal “<i>Procedure for maintenance of power meters and transformers</i>” and the VCS PD. <table><tr><th>Description</th><th>Main Meter</th><th>Back-Up Meter</th></tr><tr><td>Serial Number</td><td>MW-1509A351-02</td><td>MW-1509A352-02</td></tr><tr><td>1st Calibration date</td><td>16-December-2016</td><td>16-December-2016</td></tr><tr><td>Valid till</td><td>15-December-2019</td><td>15-December-2019</td></tr><tr><td>2nd Calibration date</td><td>15-September-2018</td><td>15-September-2018</td></tr><tr><td>Valid till</td><td>14-September-2021</td><td>14-September-2021</td></tr></table> The calibrations were held by laboratory Veritest Ltda. accredited by ONAC under accreditation code 10-LAC-032.			Description	Main Meter	Back-Up Meter	Serial Number	MW-1509A351-02	MW-1509A352-02	1st Calibration date	16-December-2016	16-December-2016	Valid till	15-December-2019	15-December-2019	2nd Calibration date	15-September-2018	15-September-2018	Valid till	14-September-2021	14-September-2021
Description	Main Meter	Back-Up Meter																			
Serial Number	MW-1509A351-02	MW-1509A352-02																			
1st Calibration date	16-December-2016	16-December-2016																			
Valid till	15-December-2019	15-December-2019																			
2nd Calibration date	15-September-2018	15-September-2018																			
Valid till	14-September-2021	14-September-2021																			

Purpose of the data	Calculation of baseline emissions
Calculation method	Not applicable
Comments	All data collected as part of the monitoring process is archived electronically and kept at least for two years after the end of the last crediting period.

4.3 Monitoring Plan

The Monitoring Plan basically consists in the procedures to meter the project's electricity generation. Since this is the only parameter to be measured and the emission factor is already set ex ante, the process is pretty straightforward, as described in the following.

Electricity generation of the project activity

Monitoring procedures are implemented onsite or remote, using tele-metering technology. The operational team is in charge of taking the measurements and reporting to XM. A main and backup meters are installed at the interconnection point of the project with the SIN₂₀. The energy meters in the substation are read via dedicated software every 24 hours. A daily report is then generated and sent to XM.

As indicated, there is a main electricity meter and a backup meter, which assures correct measurement in case of failures of the main electricity meters. If any other emergency would not allow measuring correctly the power generation of the plants or if data would be missing as a result of a failure in the monitoring process, no emission reductions shall be claimed during that period until guaranteeing again correct function of the meters and having reliable data.

The information is backed up by the operational team frequently. The data is included in an excel spreadsheet for emission reductions calculations on a monthly basis. All data collected as part of the monitoring process are archived electronically and kept at least for two years after the end of the last crediting period.

The following scheme shows the power plants, the substation and the metering points:

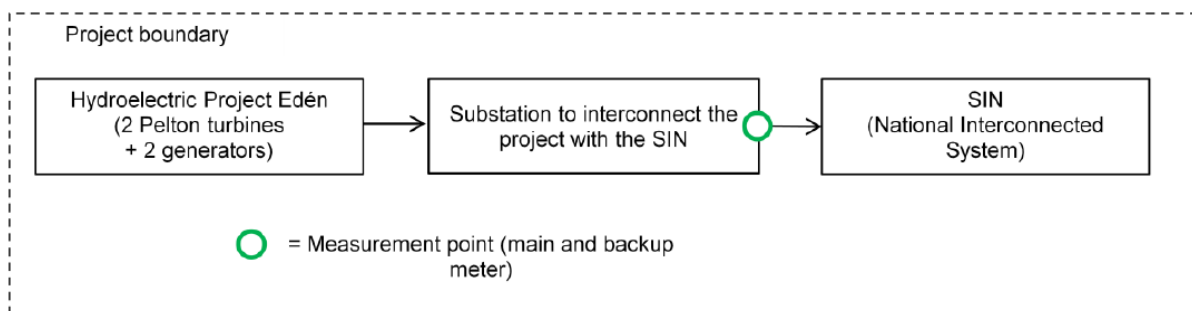


Figure 3. Simplified scheme of the monitoring.

QA/QC measures

The meters are calibrated maximum every three years. Calibration tasks follow national standards and are in accordance with the calibration instructive specified in Colombian standard NTC 4,856 for electricity metering devices. In case both meters fail, no emission reductions will be claimed during that period until having again data from the main or backup meter.

Personnel responsible for monitoring

Figure 4 shows the organizational chart of the monitoring.

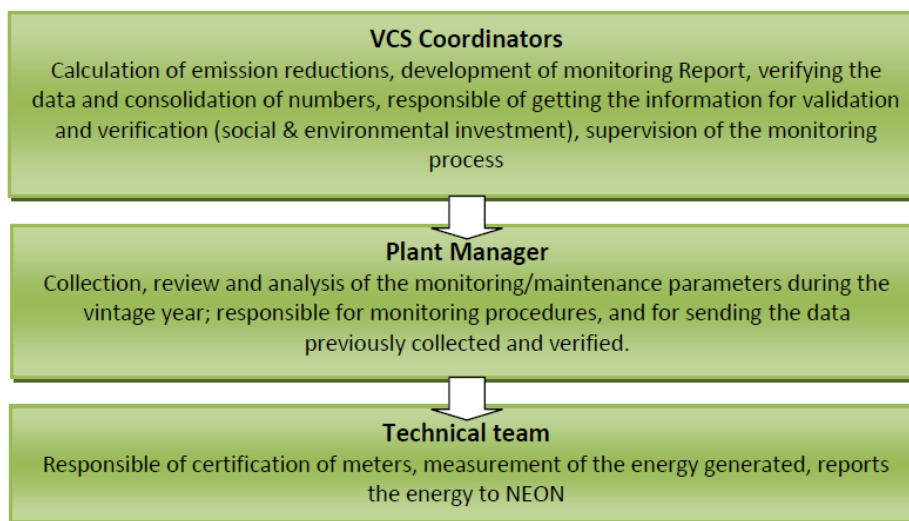


Figure 4. Operational structure of the monitoring plan

Responsible personnel:

- The VCS Coordinators supervise the monitoring process and are in charge of compiling the monitoring data in an excel spreadsheet and calculating the emission reductions of the monitoring period. They also develop the monitoring report in accordance with the VCS rules.
- The Plant Manager is responsible for verification of energy measurement. He is responsible for checking and verifying the meter readings.
- The Instrumentation and Control Engineers are responsible for electricity generation reading and for processing the energy produced by the project from the meters installed at the substation. Records of the meter are downloaded in a spreadsheet for measurement control and the data discharged from the meter is stored electronically and sent to XM (www.xm.com.co).

Personnel who carry out monitoring tasks are familiar with the basic monitoring requirements and structures. New personnel have to participate in a basic training in order to get familiarized with the monitoring procedures. Since the main monitoring tasks, i.e. the measurement of the energy production, the calibration of energy meters, and the reporting of the energy generation, are carried out independently from the VCS as part of the daily operation, no specific training is required. The VCS

Coordinators are supported by an external consultant if necessary in order to assure correct application of the monitoring procedures. They also carry out corrective actions if any inconsistency is identified and train the Plant Manager and the Technical team, if necessary.

5 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

5.1 Baseline Emissions

The baseline scenario and the emission rate calculation are based on the electricity that otherwise would have been generated by the plants connected to the grid and by addition of future plants. According to the methodology, no leakage emissions are considered.

Baseline emissions are calculated by multiplying the combined margin emission factor ($EF_{grid,y}$, in tCO_2e/MWh) by the electricity generated by the proposed project activity during the year y (EG_y , in MWh).

Baseline emissions include only CO_2 emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation by the project activity would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants.

The detailed calculations and data of the baseline emissions are presented in the Excel file “El Edén_ER Calculation Verification”, for which the following formulas are presented.

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	=	Baseline emissions in year y (tCO_2/yr)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)
$EF_{grid,y}$	=	Combined margin CO_2 emission factor for grid connected power generation in year y calculated using version 3.0.0 of the “Tool to calculate the emission factor for an electricity system” (tCO_2/MWh)

Since it is a green-field project, according to Eq. (7) of ACM0002,

$$EG_{PJ,y} = EG_{facility,y}$$

The expression $EG_{PJ,y}$ was used in the VCS PD and it is also used in this Monitoring Report. This parameter was measured during the monitoring period as established in the VCS PD. For more detail, please refer to the file: “El Edén_ER Calculation Verification”

The combined margin emission factor ($EF_{grid,CM,y}$) was calculated according to the guidance in the “Tool to calculate the emission factor for an electricity system” (version 3.0.0) in the VCS PD and set as a fix parameter for this MR. Hence the value applied was 0.3072 tCO₂/MWh.

The baseline emissions are calculated as shown in the following table:

From	To	Ex Post	Ex Ante	BE _y (tCO ₂)
		EG _{facility,y}	EF _{grid,CM,y}	
		(MWh/yr)	(tCO ₂ /MWh)	
16/02/2017	31/12/2017	66,783.62	0.3072	20,515
1/01/2018	31/12/2018	78,259.22	0.3072	24,041
1/01/2019	31/12/2019	48,055.58	0.3072	14,762
1/01/2020	31/12/2020	44,898.41	0.3072	13,792
16/02/2017	31/12/2020	237,996.83	0.3072	73,110

5.2 Project Emissions

As per the applied ACM0002 version 13.0.0, for hydro power project activities that result in new reservoirs, project proponents do not need to account for CH₄ and CO₂ emissions from the reservoir, if the power density of the project activity (PD) is greater than 10 W/m².

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$

Where:

- PD = Power density of the project activity (W/m²).
- Cap_{PJ} = Installed capacity of the hydro power plant after the implementation of the project activity (W)
- Cap_{BL} = Installed capacity of the hydro power plant before the implementation of the project activity (W). For new hydro power plants, this value is zero
- A_{PJ} = Area of the reservoir measured in the surface of the water, after the implementation of the project activity, when the reservoir is full (m²)
- A_{BL} = Area of the reservoir measured in the surface of the water, before the implementation of the project activity, when the reservoir is full (m²). For new reservoirs, this value is zero.

Since the project activity has no reservoir, this step is not required, and the project emissions are zero:

$$PE_y = 0$$

5.3 Leakage

No leakage emissions are considered as provided in the methodology:

LEy = 0

5.4 Net GHG Emission Reductions and Removals

Year	Baseline emissions or removals (tCO ₂ e)	Project emissions or removals (tCO ₂ e)	Leakage emissions (tCO ₂ e)	Net GHG emission reductions or removals (tCO ₂ e)
2017 ⁵	20,515	-	-	20,515
2018	24,041	-	-	24,041
2019	14,762	-	-	14,762
2020	13,792	-	-	13,792
Total	73,110	-	-	73,110

The following table shows the difference between the ex-ante estimation of GHG emission reductions presented in the VCS PD and the actual reductions based on the monitored parameters. There is a noticeable difference between the two values, since ex-ante ERs are higher than the reductions presented on this Monitoring Report. This could be attributed to the fact that estimation of electricity generation at a validation stage of a green-field plant are usually based on historical flow rates that don't necessarily represent reality.

Furthermore, even though there have been no mayor breakdowns during the operation of the hydroelectric project, electricity generation has been affected during periods of electrical storms or mechanical issues that have caused some short stops during the operation. There were also, some short periods when the river flow was too low, thus reducing the electricity generation or in some cases, causing short stops.

Other partial but longer stops of the equipment have taken place for routine maintenance or issues related to the electrical equipment that have been quickly addressed and operation restored.

⁵ From 16-February-2017 to 31/December/2017

Item		Values estimated in ex-ante calculation of registered PDD	Actual values achieved during this monitoring period
Year 2017	From	16/02/2017	16/02/2017
	To	31/12/2017	31/12/2017
	Days	319	319
	Emission reductions or GHG removals by sinks (t CO ₂ e)	28,056	20,515
	tCO ₂ e/day	88	64
Year 2018	From	1/01/2018	1/01/2018
	To	31/12/2018	31/12/2018
	Days	365	365
	Emission reductions or GHG removals by sinks (t CO ₂ e)	32,102	24,041
	tCO ₂ e/day	88	66
Year 2019	From	1/01/2019	1/01/2019
	To	31/12/2019	31/12/2019
	Days	365	365
	Emission reductions or GHG removals by sinks (t CO ₂ e)	32,102	14,762
	tCO ₂ e/day	88	40
Year 2020	From	1/01/2020	1/01/2020
	To	31/12/2020	31/12/2020
	Days	366	366
	Emission reductions or GHG removals by sinks (t CO ₂ e)	32,190	13,792
	tCO ₂ e/day	88	38